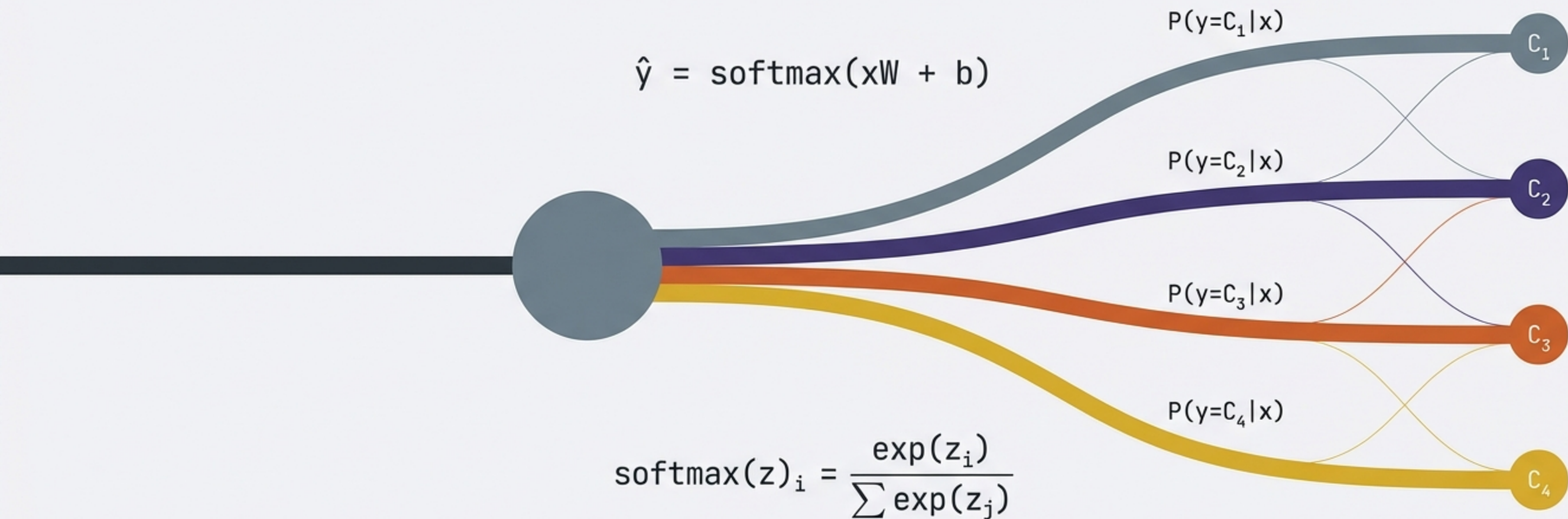


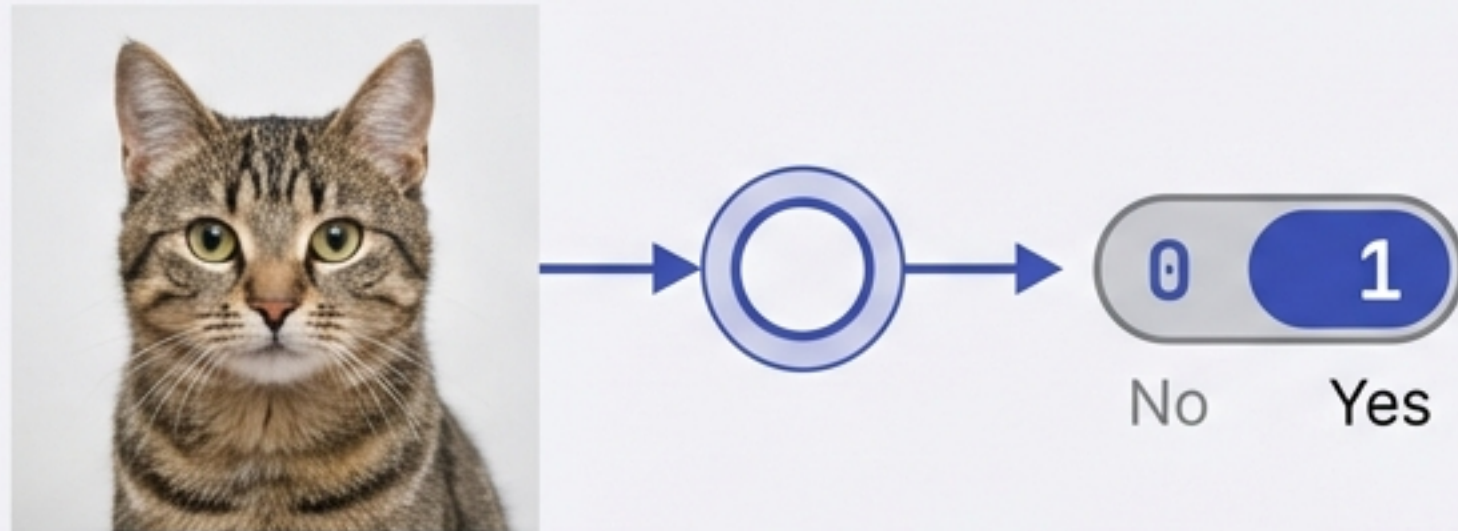
Softmax Regression

Generalising Logistic Regression for Multi-Class Classification



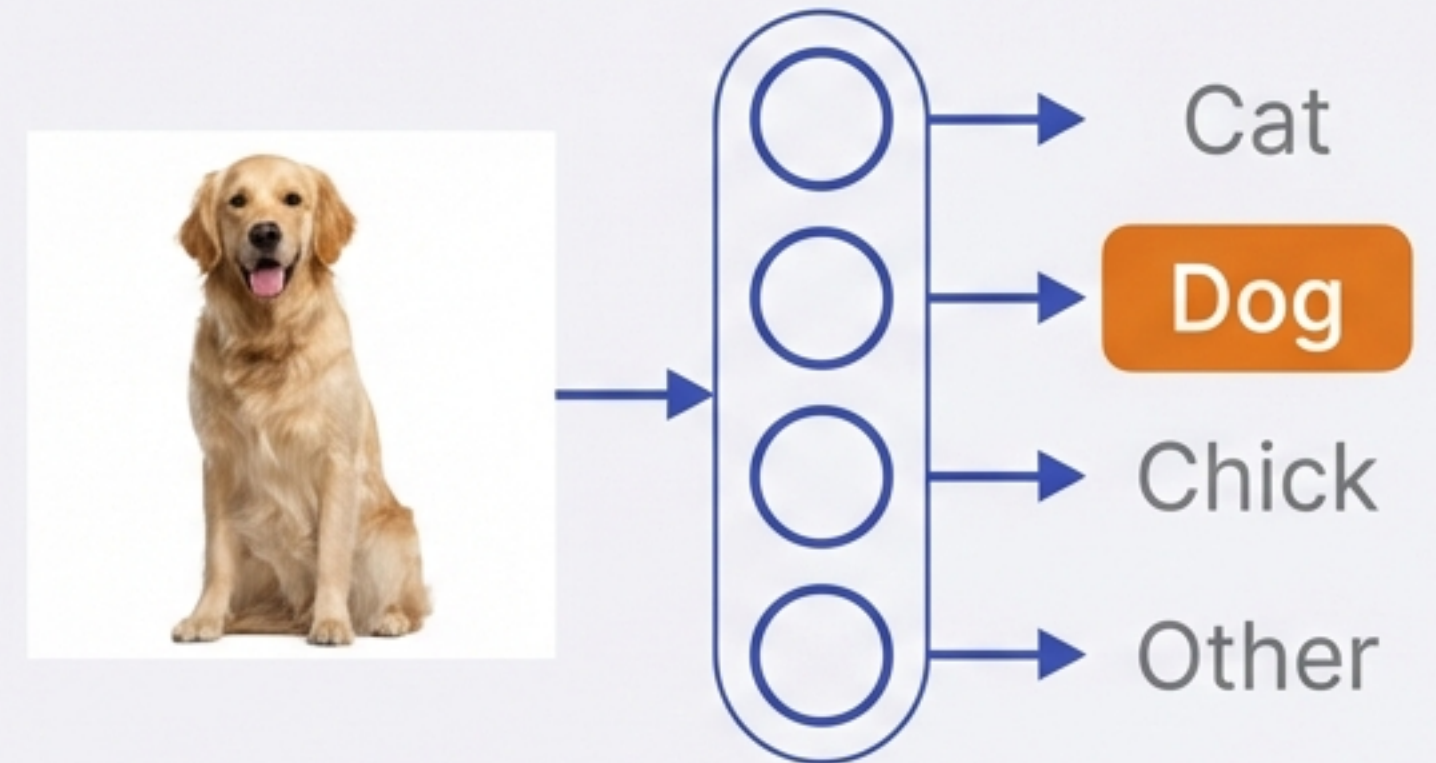
Moving Beyond Binary Classification

Logistic Regression



Is this a cat?
(Yes/No)

Softmax Regression

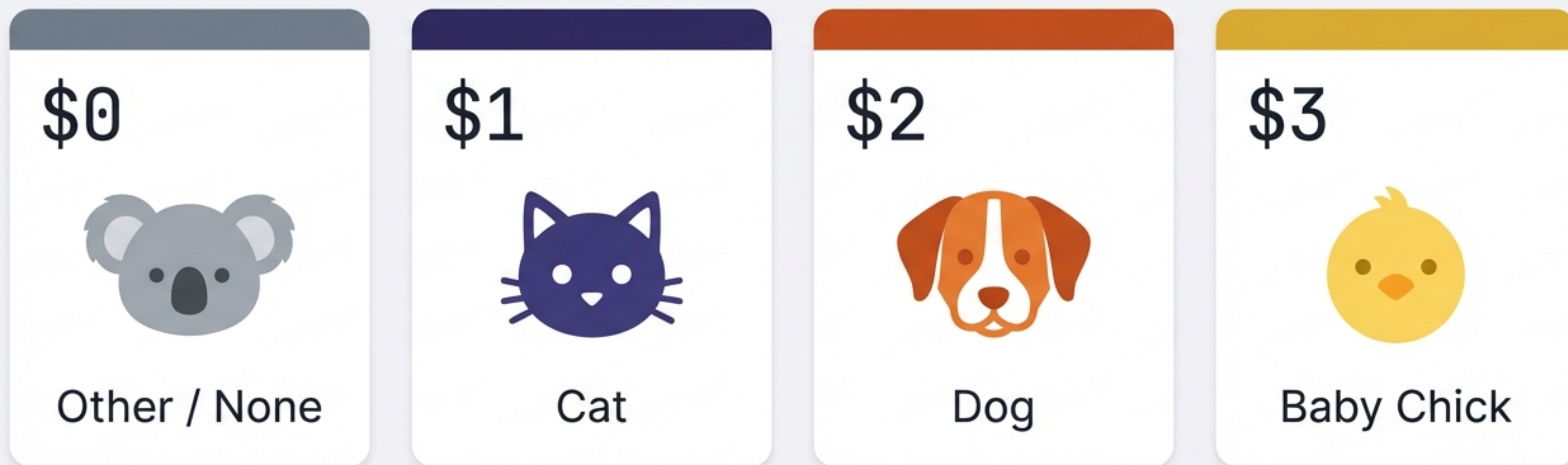


Which *one* of these
 C classes is this?

Real-world problems rarely represent a simple binary choice.
We need a probability distribution over C possible classes.

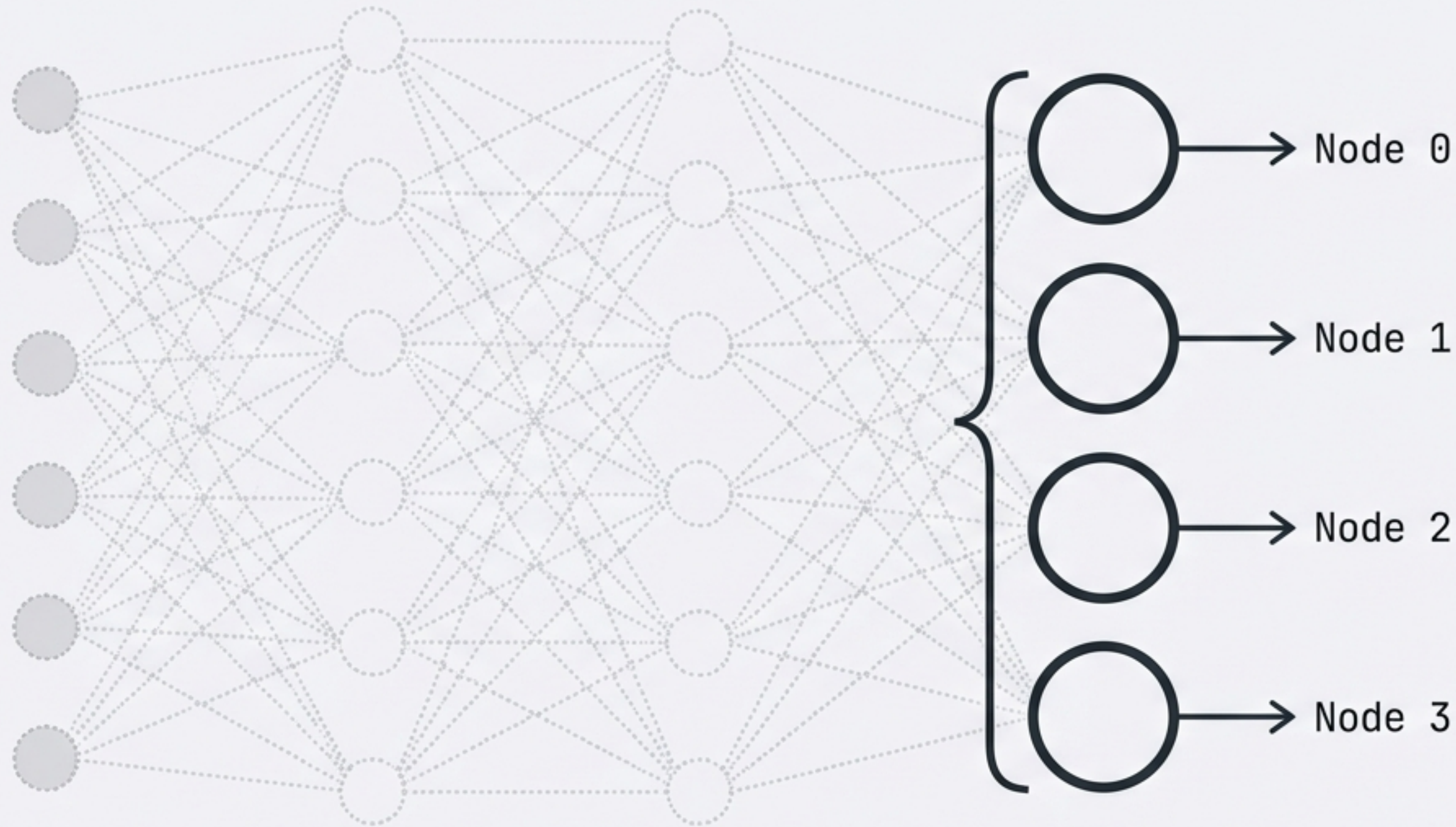
Defining the Scope: C Classes

In this deck, we assume $C = 4$.



Labels range from \$0 to $C-1$

Architecture of the Output Layer



$$n^{[L]} = C = 4 \text{ units}$$

Unlike binary classification which outputs a single number, Softmax requires a separate node for every possible class. Each node is responsible for estimating the probability of one specific category.

The Goal: A Probability Distribution

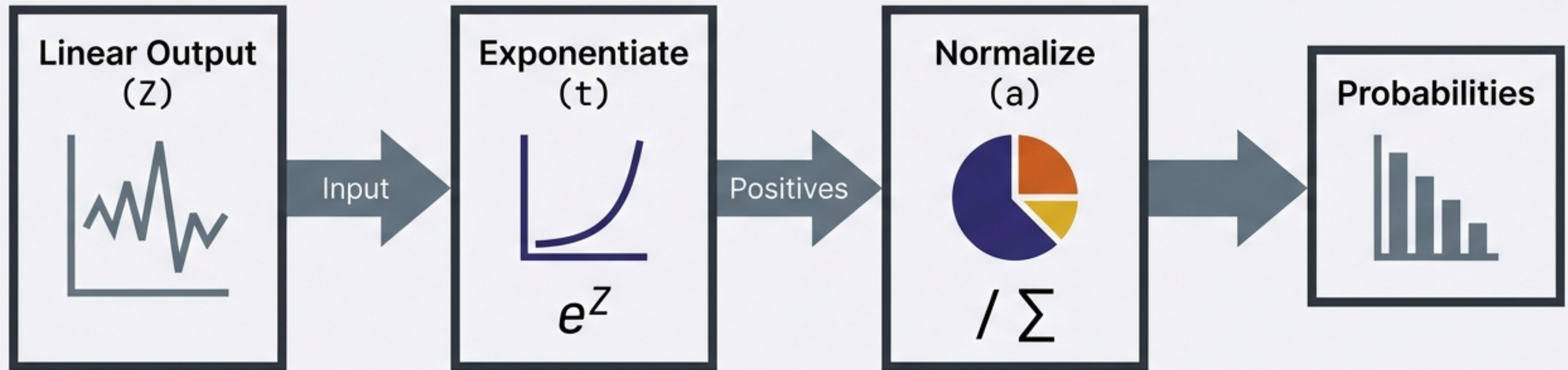
$$\begin{bmatrix} P(\text{Other}|x) \\ P(\text{Cat}|x) \\ P(\text{Dog}|x) \\ P(\text{Chick}|x) \end{bmatrix} \hat{\mathbf{y}} \text{ or } \mathbf{a}^{[L]}$$

$$\sum \hat{y}_i = 1$$

The sum of all elements in the output vector must equal exactly 1 (100%).

The Softmax Activation Function

Transforming raw scores into valid probabilities.



The Softmax layer applies a unique sequence of operations to ensure the output vector represents a valid probability distribution.

Step 1: The Linear Output ($Z^{[L]}$)

First, the layer computes the raw linear combination (Logits).

$$Z^{[L]} = W^{[L]}a^{[L-1]} + b^{[L]}$$



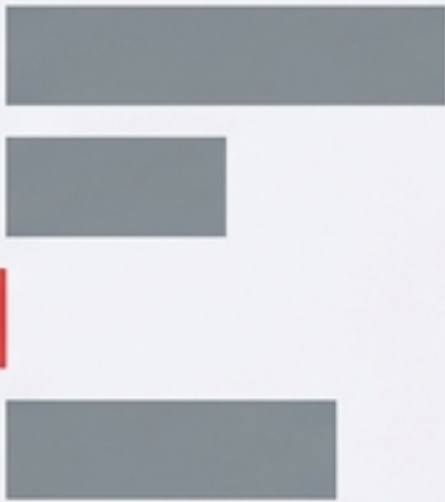
Note: Raw scores can be negative, positive, and of any magnitude. They are not yet probabilities.

Step 2: Exponentiation

Before

z

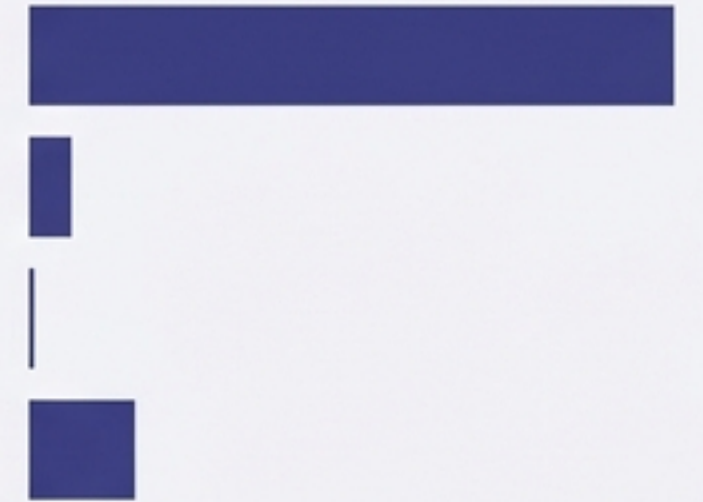
$$\begin{bmatrix} 5 \\ 2 \\ -1 \\ 3 \end{bmatrix}$$



After

t

$$\begin{bmatrix} e^5 \approx \mathbf{148.4} \\ e^2 \approx 7.4 \\ e^{-1} \approx 0.4 \\ e^3 \approx 20.1 \end{bmatrix}$$



Exponentiation makes all values **positive** and exaggerates differences. The gap between 5 and 2 is small; the gap between 148.4 and 7.4 is massive.

Step 3: Normalisation

$$\sum [148.4, 7.4, 0.4, 20.1]$$

Total Sum = 176.3

0:	148.4	/	176.3	=	0.842
1:	7.4	/	176.3	=	0.042
2:	0.4	/	176.3	=	0.002
3:	20.1	/	176.3	=	0.114

Class 0 Prediction: 84.2%

$$\mathbf{a}^{[L]} = [0.842, 0.042, 0.002, 0.114]$$

Sum = 1.0

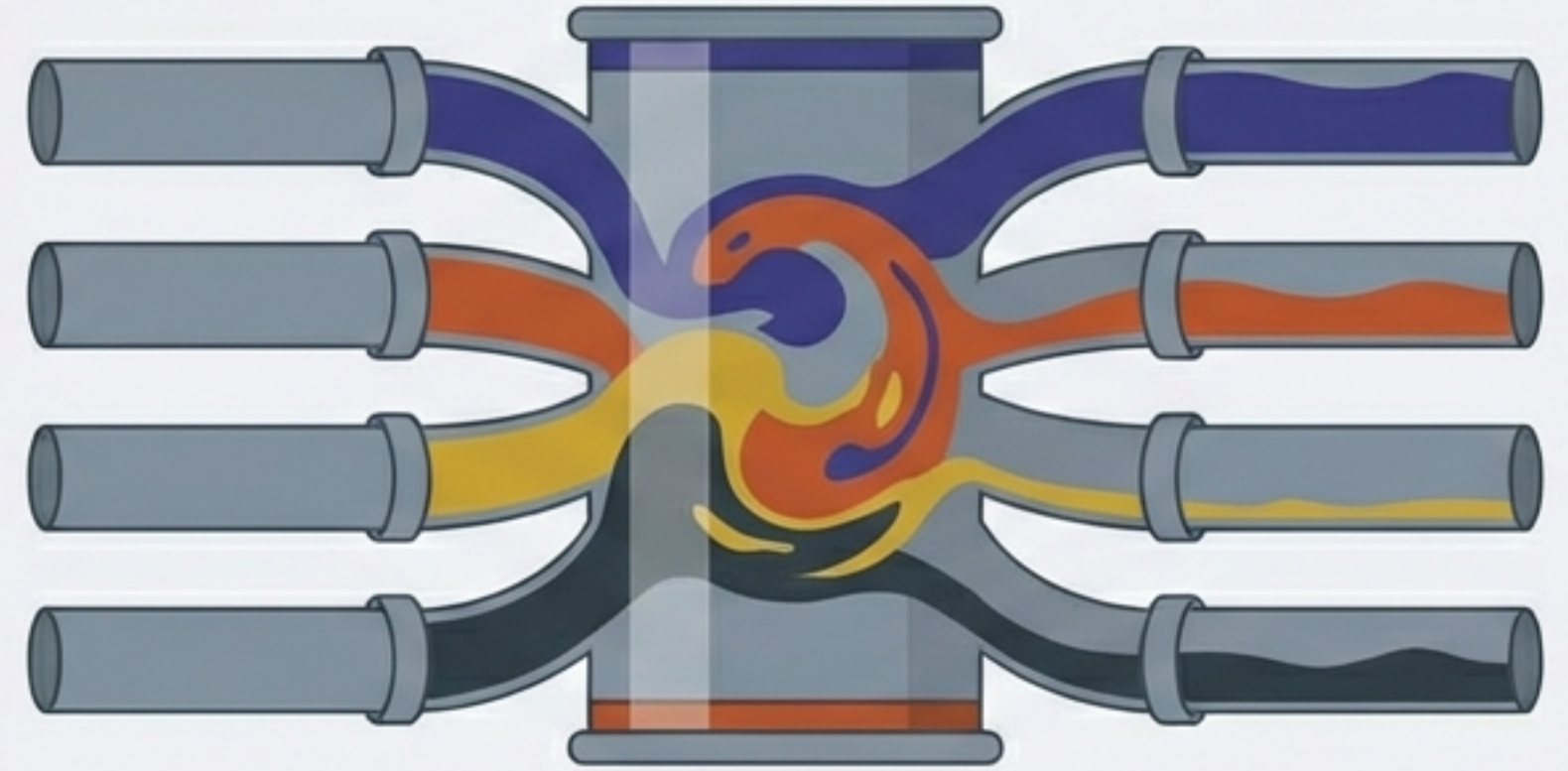
Softmax is a Vector Activation Function

Standard Activation - ReLU/Sigmoid



Scalar in, Scalar out. Independent.

Softmax Activation

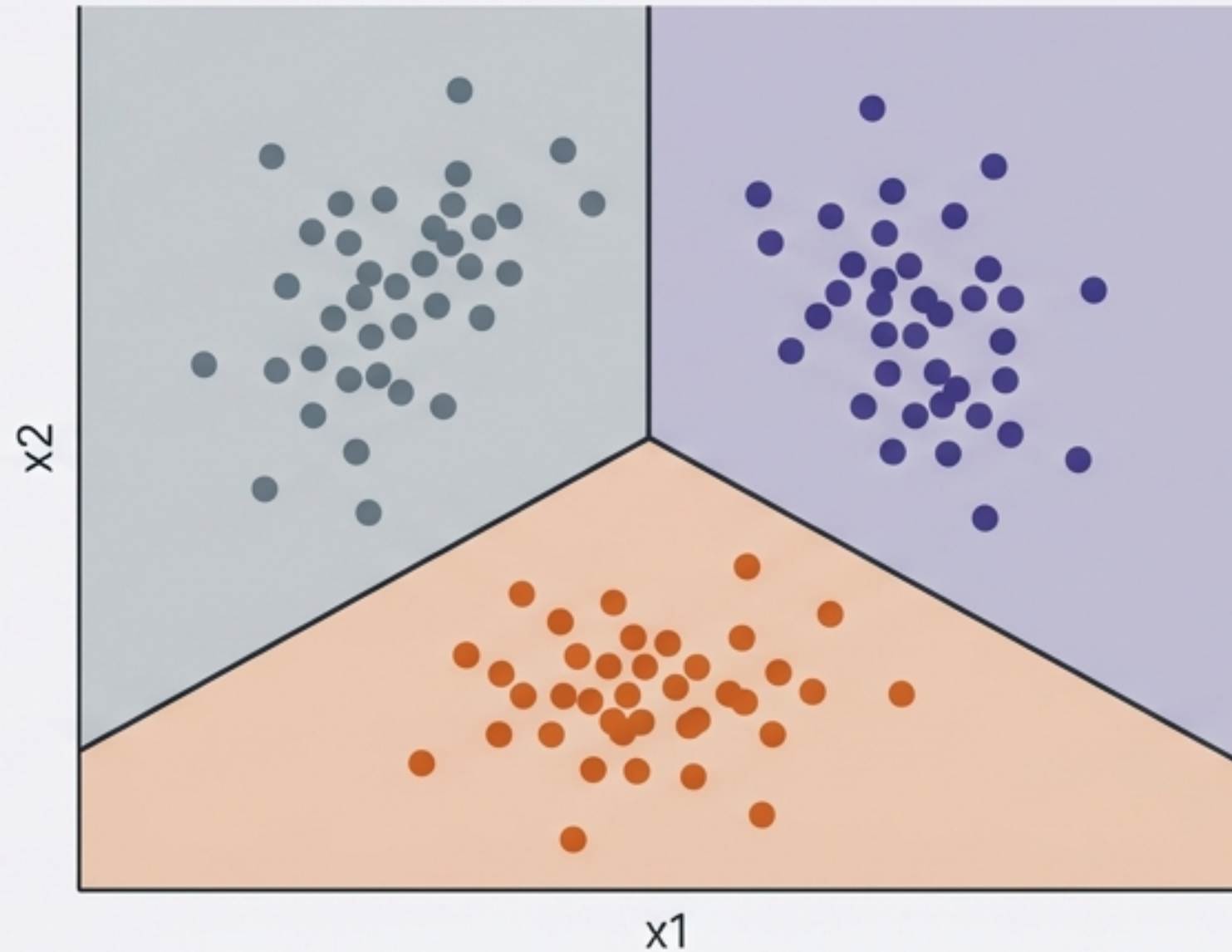


Vector in, Vector out. Cross-dependent.

Unlike Sigmoid or ReLU, Softmax is cross-dependent. The probability of 'Cat' depends on the scores of 'Dog' and 'Chick' due to the shared normalisation denominator.

Intuition: Linear Decision Boundaries

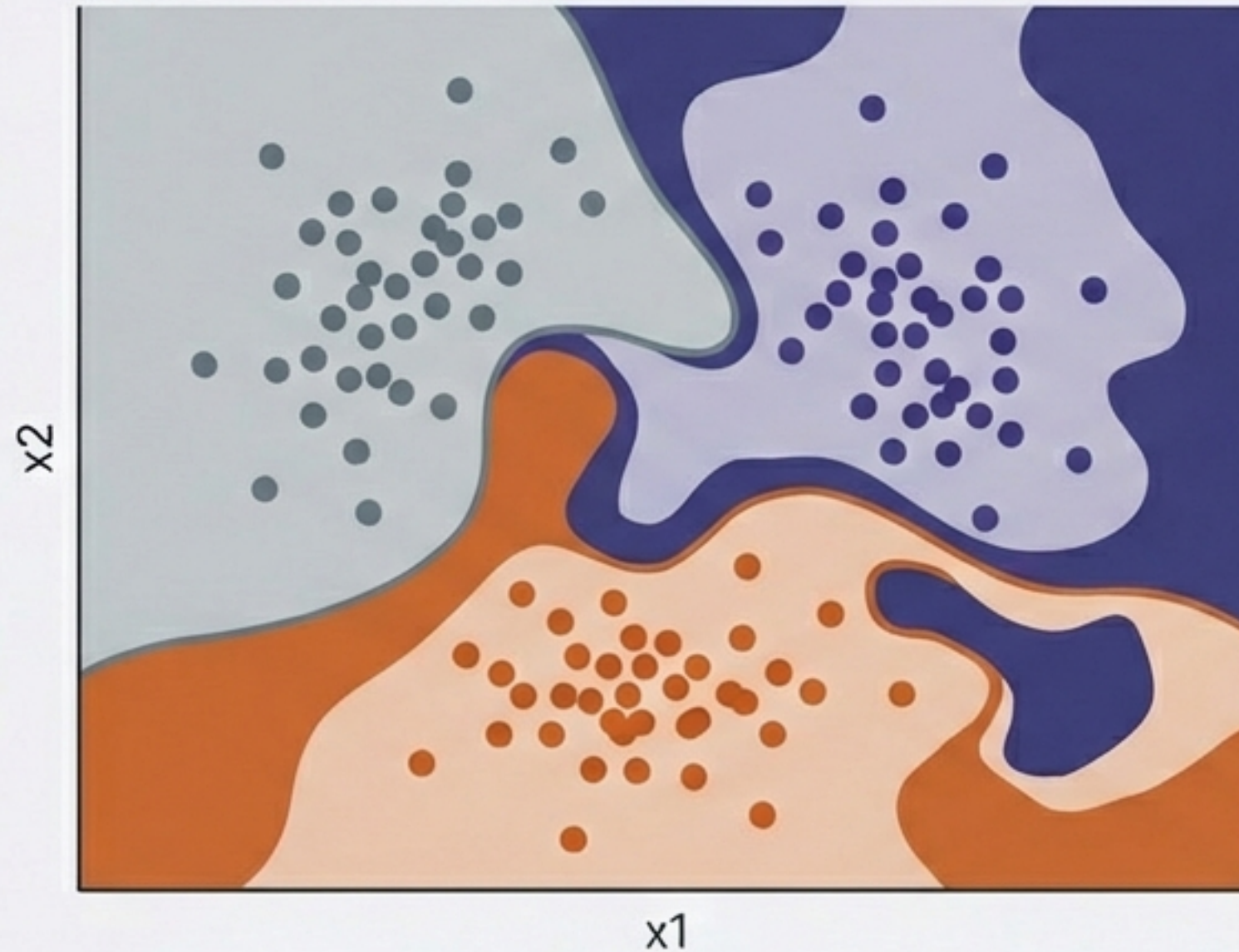
Softmax without hidden layers.



Much like Logistic Regression creates a linear split between 2 classes, simple Softmax generalizes this to create linear boundaries between C classes.

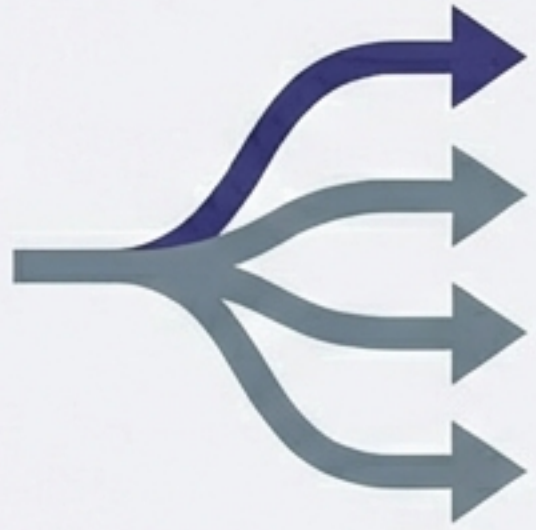
Deep Networks Create Complex Boundaries

Softmax *with* hidden layers.

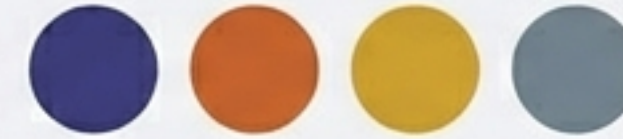


Hidden layers learn complex features and distort the input space. The final Softmax layer acts on these transformed features, allowing for highly non-linear classifications.

Key Takeaways



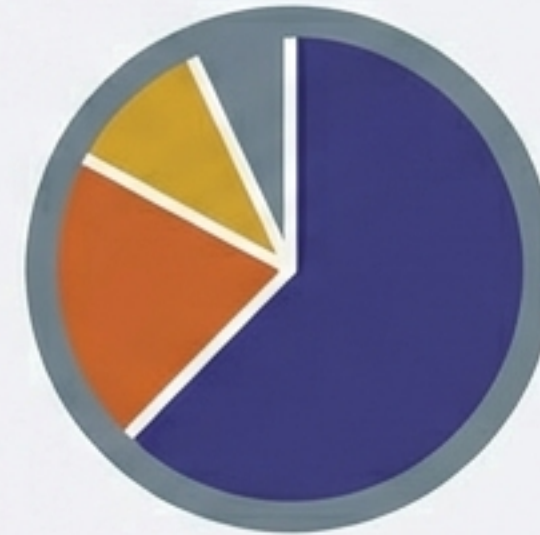
Generalises
Logistic Regression
to C classes.



Output layer
requires C
distinct units.

$$e^z / \Sigma$$

Exponentiate to
ensure positivity,
Normalize to
ensure probability.



Outputs a valid
distribution
summing to 1.0.